

Hayato Isa

Seattle, WA | 510-365-3556 | iamhayatoisa@berkeley.edu | github.com/hayathorium | hayathorium.github.io

EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Arts in Applied Mathematics and Data Science

May. 2026

- GPA: 3.89/4.00, Cum Laude
- **Relevant Coursework:** Bayesian Inference & Hypothesis Testing (DATA C102), Probability Theory (STAT 134), Data Cleaning & Visualization (DATA C100), Machine Learning (CS 188), Data Privacy & Ethics (DATA C104)

PROJECTS & EXPERIENCE

AI Conversation Analysis | *HTML/CSS, JavaScript, Node.js, Gemini API, Google Cloud, PM2, Linux* Jul. 2026

- Developed a full-stack dashboard that automatically scrapes Discord group chat history and leverages the Gemini API to profile detailed user personalities and interpersonal relationship dynamics.
- Engineered a GDPR-compliant simulation environment on Google Cloud Linux VMs, deploying 20 autonomous Gemini AI agents to generate synthetic chat data for testing and demonstration.
- Utilized PM2 process management to guarantee 24/7 system uptime, implementing automated recovery protocols to seamlessly handle runtime failures.

Flower Image Generator | *Python, PyTorch, Deep Learning, Computer Vision, GANs* Jun. 2026

- Implemented a Deep Convolutional Generative Adversarial Network (DCGAN) from scratch in PyTorch to synthesize 64x64 resolution images using the Oxford 102 Flowers dataset.
- Built an end-to-end training pipeline including dataset preprocessing, model persistence, automated image generation, and loss monitoring, enabling effortless iterative model optimization.

COPD Prevalence Analysis | *Bayesian Inference, PyMC, Altair, Data Cleaning, Data Interpretation* Dec. 2025

- Developed a Bayesian hierarchical model using PyMC to quantify state-level COPD risk factors, achieving a 28% reduction in RMSE (0.56 vs. 0.78 OLS baseline) by integrating latent state effects with diverse public health data.
- Engineered a robust end-to-end data pipeline to clean and merge complex datasets, conducting MCMC-based posterior inference and sensitivity analysis to validate model outcomes before communicating findings through interactive Altair visualizations.

Higgs Boson Discovery Replication Study | *Python, Hypothesis Testing, Data Filtering* Dec. 2025

- Replicated the ATLAS Higgs boson discovery analysis using CERN Open Data by implementing event selection, data filtering, and a reproducible analysis pipeline based on the original research methodology.
- Performed statistical hypothesis testing using the profile likelihood ratio to detect the Higgs boson signal, achieving a statistical significance of $p \approx 2.05 \times 10^{-3}$ while documenting all preprocessing and validation steps for reproducibility.

CIFAR-10 Image Classification System | *Python, PyTorch, CNN, ResNet-34, Torchvision* Jun. 2026

- Built and trained a custom Convolutional Neural Network in PyTorch from scratch for 10-class image classification to develop a foundational understanding of CNN architectures.
- Implemented model checkpointing for training persistence and extended the pipeline with a ResNet-34 architecture to enhance classification performance and training stability.

Secure File Sharing System | *Go, Cryptography, PKI, HMAC, AES* Apr. 2026

- Designed and implemented a cryptographically secure file storage and sharing system in Go, supporting user authentication, file operations, and access control over an untrusted datastore adversary.
- Applied public-key encryption (PKE), digital signatures, HMAC, and symmetric-key encryption (AES) to guarantee confidentiality and integrity of all user data, even against a malicious server with full read/write access.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, C++, Go, MATLAB, HTML/CSS, $\text{\LaTeX}$

Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, CNNs, GANs, ResNet

Statistical Modeling: Bayesian Inference, PyMC, Hypothesis Testing, Regression, Feature Engineering

Data Analysis: pandas, NumPy, SciPy, Altair, Matplotlib, Seaborn

Backend: Node.js, Discord.js, REST APIs, Authentication, File Systems, PM2

Tools: Google Cloud, Google Colab, Linux, Git, Jupyter, VS Code, IntelliJ IDEA, Excel